GoodTimers: A Walk Planning Application

Individual Experiment Report

MS-IV | CPSC 444

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1. Introduction

University students regularly experience unstructured downtime between classes, yet few digital tools exist to help them make spontaneous, personalized use of that time. At the University of British Columbia (UBC), where the campus spans a large, walkable geography, students frequently miss opportunities to take intentional, restorative breaks. Prior HCI research has explored leisure and activity recommendation systems, as well as novel mobile interaction paradigms for rapid data collection, but no work has specifically addressed on-campus break planning through a route-generation interface tailored to student preferences.

This project, GoodTimers, responds to that gap. Our team designed a mobile application prototype that generates a walking route around UBC based on a set of user-specified decision factors, including time availability, willingness to walk, desired sociability, and destination type. The core design challenge we investigated was the mechanism through which users input these decision factors. We developed and evaluated three data collection approaches (DCAs): a standard survey using sliders, a swipe-based card interface, and a novel drag interaction in which the user draws a single continuous gesture through all questions simultaneously.

This report describes the formal experiment we conducted to evaluate our Proposed Design Solution (PDS) against a Baseline Planning Approach (BPA) consisting of standard Google Maps usage. The experiment was designed to test four hypotheses:

Hypothesis	**Statement**
H1	Self-rated cognitive load is lower when planning a walk using the PDS compared to the BPA.
H2	The reduction in cognitive load between BPA and PDS is similar across all three DCA variants (no interaction effect).
H3	The PDS map output is more readable and/or more easily editable than the BPA map (Google Maps).
H4	One or more DCA variants differs from the others in task completion time and/or number of input errors.

2. Experiment Description

2.1 Participants

Twelve (N = 12) participants were recruited through convenience sampling within the UBC student community. All participants were current UBC students, reflecting the application's target demographic. Participants were assigned to one of three DCA conditions (Drag, Survey, Swipe), with four participants per condition. Informed consent was obtained from all participants prior to the study.

2.2 Variables and Measures

The experiment employed the following independent variables (IVs) and dependent variables (DVs):

Variable	**Type**	**Levels / Measure**
Task (Planning Approach)	IV – Within-subjects	BPA (Google Maps) vs. PDS (prototype)
DCA (Decision Collection Approach)	IV – Between-subjects	Drag, Survey, Swipe
Task Order	IV – Counterbalancing factor	BPA-first vs. PDS-first
Self-rated cognitive load	DV (H1, H2)	Single 7-point Likert item post-task
Map satisfaction	DV (H3)	Single 7-point Likert item post-task
Map read time	DV (H3)	Seconds, measured by experimenter
DCA planning time	DV (H4)	Seconds, measured by experimenter

2.3 Design

The experiment used a 2 x 3 mixed factorial design. The within-subjects factor was Task (BPA vs. PDS), so every participant completed both planning scenarios. The between-subjects factor was DCA (Drag, Survey, or Swipe), with each participant assigned to only one DCA variant for the PDS condition. Task order was counterbalanced across participants to control for learning and fatigue effects. The result was a design that allowed us to detect both a main effect of the interface (H1) and potential interaction effects between the interface and DCA type (H2).

2.4 Procedure

Each session was conducted in person by two members of the research team: one facilitator and one note-taker. The procedure proceeded as follows:

- Consent and briefing: Participants were given an overview of the study and signed an informed consent form. They were told they would be planning a walk around campus using two different tools.
- Scenario delivery: Participants received a written user scenario describing their context (e.g., a 30-minute break between classes, wanting a moderately social walk).
- Task A and Task B: Each participant completed both tasks in their assigned order. For the BPA task, participants used Google Maps on a provided device to plan a walk. For the PDS task, they used the GoodTimers prototype assigned to their DCA condition. For the Drag condition only, a brief demonstration of the drag mechanism was provided immediately before the PDS task, based on a finding from the pilot study (see Section 2.6).
- Post-task questionnaires: After each task, participants completed a short Qualtrics questionnaire including a cognitive load item and a map satisfaction item.
- Semi-structured interview: Following both tasks, participants took part in a brief semi-structured interview to capture qualitative feedback on both the BPA and PDS experiences, the DCA they used, and whether they would use the app in the future.

2.5 Tasks

Task A (BPA): Participants were asked to use Google Maps to plan a 30-minute walking route around UBC campus based on the provided scenario. They were then asked to identify the route's key stops and describe how they found the planning experience.

Task B (PDS): Participants used the GoodTimers prototype to input their decision factors using their assigned DCA (Drag, Survey, or Swipe), waited through a simulated loading screen, and then evaluated the generated map output. The map used in this task was a custom-designed static map of UBC showing a walking route with marked stops and a directional arrow – distinct from the Google Maps interface used in Task A.

2.6 Divergences from the Prototype Phase (MSIII)

Several adjustments were made to the experimental protocol following the pilot study:

- Elimination of the 'update walking route' subtask: The original protocol asked participants to revise the route based on an updated scenario. This was removed as redundant – the ability to input decision factors was already assessed through the primary PDS task.
- Addition of a drag demonstration: Due to difficulty observed during the pilot with the drag interaction, participants assigned to the Drag DCA were given a brief demonstrative trial of the mechanism before beginning Task B.
- Expansion of qualitative data collection: Based on TA feedback during the pilot debrief, a post-experiment semi-structured interview was added to supplement the single open-ended question that had originally appeared in each post-task questionnaire.

3. Results

Results are presented per hypothesis. All statistical analyses were conducted in R. Due to the small sample size and non-normal distributions, non-parametric or rank-based tests were prioritized where appropriate.

3.1 H1 and H2: Cognitive Load and DCA Interaction Effect

To test H1 and H2 simultaneously, an Aligned Rank Transform (ART) ANOVA was applied to the self-rated cognitive load scores, with Task, DCA, and Task Order as factors. The ART procedure is appropriate for factorial designs with ordinal dependent variables where parametric assumptions may not hold.

H1 was supported: a statistically significant main effect of Task was found ($F(1, 6) = 13.47$, $p = 0.010$), with participants reporting lower cognitive load when using the PDS compared to the BPA. The partial eta-squared for Task was 0.69, indicating a large practical effect.

H2 was also supported: no statistically significant interaction effect was found between Task and DCA ($F(2, 6) = 1.78$, $p = 0.247$), suggesting that the cognitive load reduction provided by the PDS was consistent across all three DCA variants. Notably, a significant interaction was found between Task and Task Order ($F(1, 6) = 8.82$, $p = 0.025$), as well as between DCA and Task Order ($F(2, 6) = 5.15$, $p = 0.050$), indicating that the order in which participants completed the two tasks influenced their reported cognitive load – a carryover effect worth noting as a limitation.

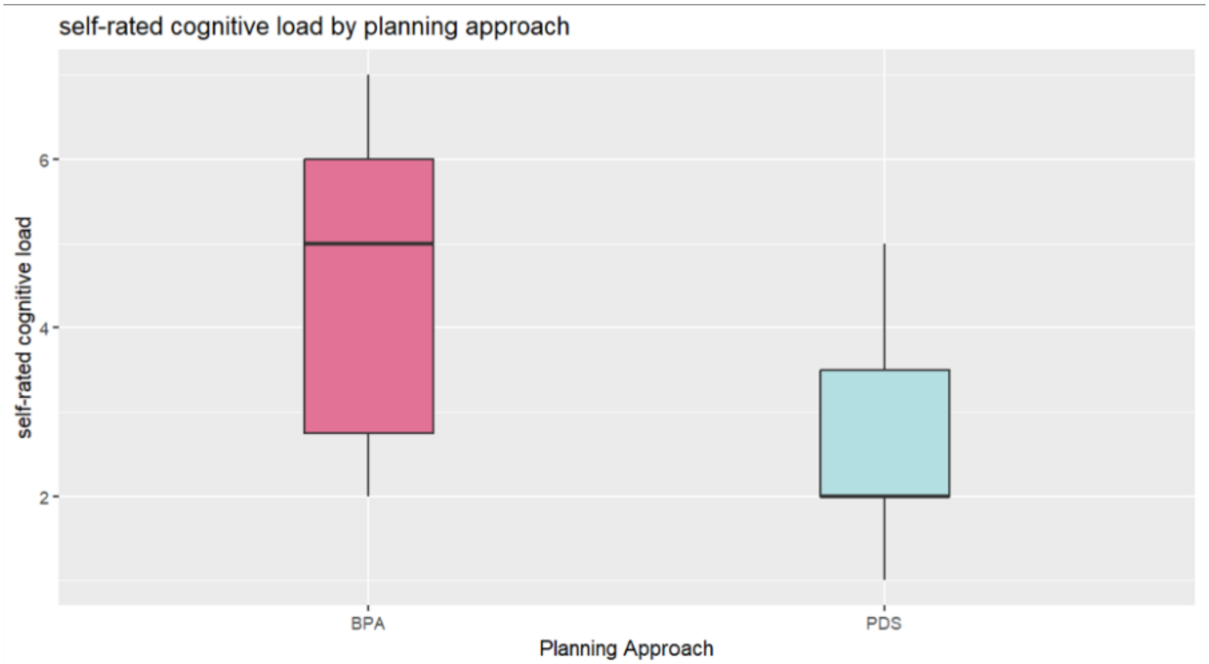

Figure 1.1: Distribution of self-rated cognitive load for BPA vs PDS. Lower scores indicate less cognitive effort.

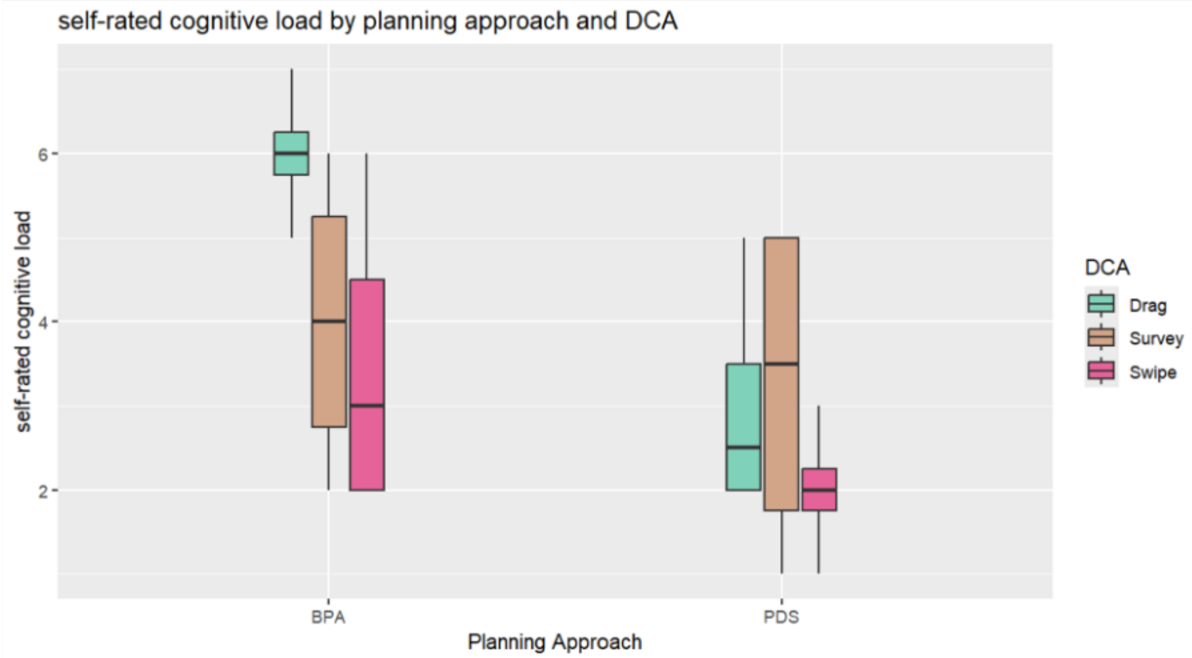

Figure 1.2: Cognitive load by planning approach, further split by DCA condition. No significant interaction effect was found (p = 0.247).

Figure 1.3: Cognitive load by planning approach, conditioned by task order. A significant interaction was observed (p = 0.025).

3.2 H3: Map Readability and Satisfaction

H3 was tested across two dependent variables: map satisfaction (Likert) and map read time (seconds).

For map satisfaction, a paired Wilcoxon signed-rank test revealed no statistically significant difference between the BPA and PDS maps ($V = 17.5$, $p = 1.0$). H3 was therefore not supported on the satisfaction measure.

For map read time, a paired one-tailed t-test found that the BPA map (Google Maps) was read significantly faster than the PDS map ($t(11) = -2.925$, $p = 0.007$), with a mean difference of -2.38 seconds. Hedge's g was -1.04, indicating a large practical effect. This result is directionally opposite to what H3 predicted: rather than the PDS map being more readable, the BPA map was faster to read. H3 was therefore not supported and was in fact contradicted for the read time measure.

Test	Statistic	p-value	Effect Size	Interpretation
Wilcoxon (map satisfaction)	V = 17.5	p = 1.0	—	No significant difference
Paired t-test (map read time)	t(11) = -2.925	p = 0.007	Hedge's g = -1.04 (large)	BPA map read significantly faster

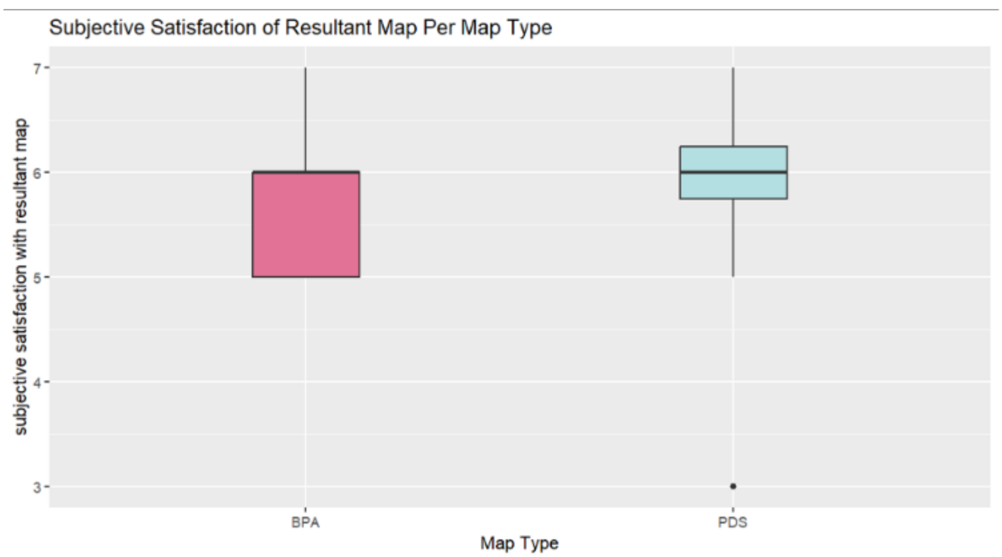

Figure 2.1: Distribution of map satisfaction scores for BPA and PDS. No significant difference was found (p = 1.0).

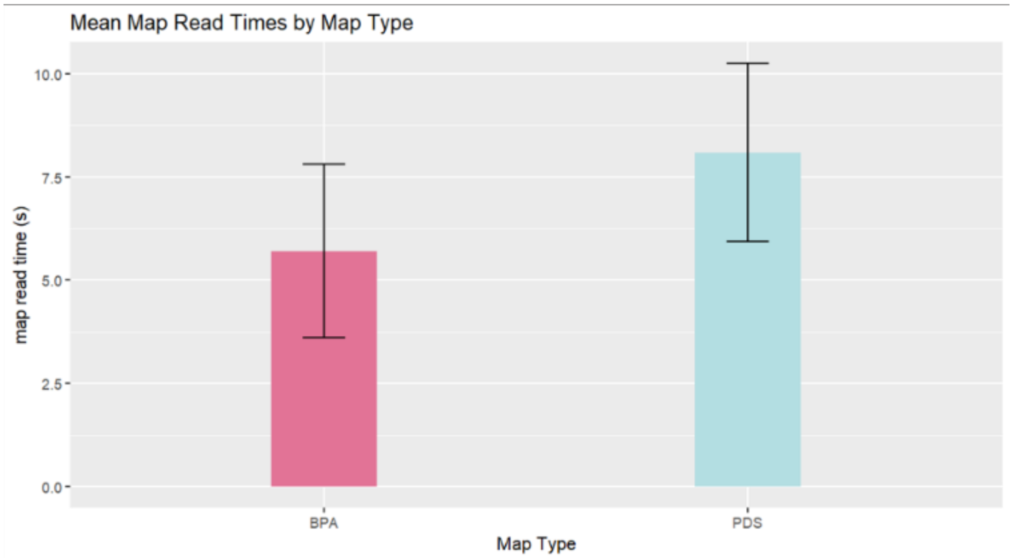

Figure 2.2: Mean map read times with standard deviation error bars. BPA was read significantly faster than PDS (p = 0.007, Hedge's g = -1.04).

3.3 H4: DCA Completion Time Differences

A Kruskal-Wallis rank-sum test was used to compare DCA planning times across the three conditions (Drag, Survey, Swipe). A one-way ANOVA was not conducted because distributional assumptions were violated: the Drag condition showed marked right-skew with a high outlier, while the Survey and Swipe distributions appeared approximately symmetric. The Kruskal-Wallis test found no statistically significant difference across DCA types ($\chi^2(2) = 0.154$, $p = 0.926$). H4 was not supported.

Visual inspection of the boxplot (Figure 3.1) reveals, however, that the Drag condition had notably higher variance and a substantial outlier, consistent with participant reports of difficulty using that interaction.

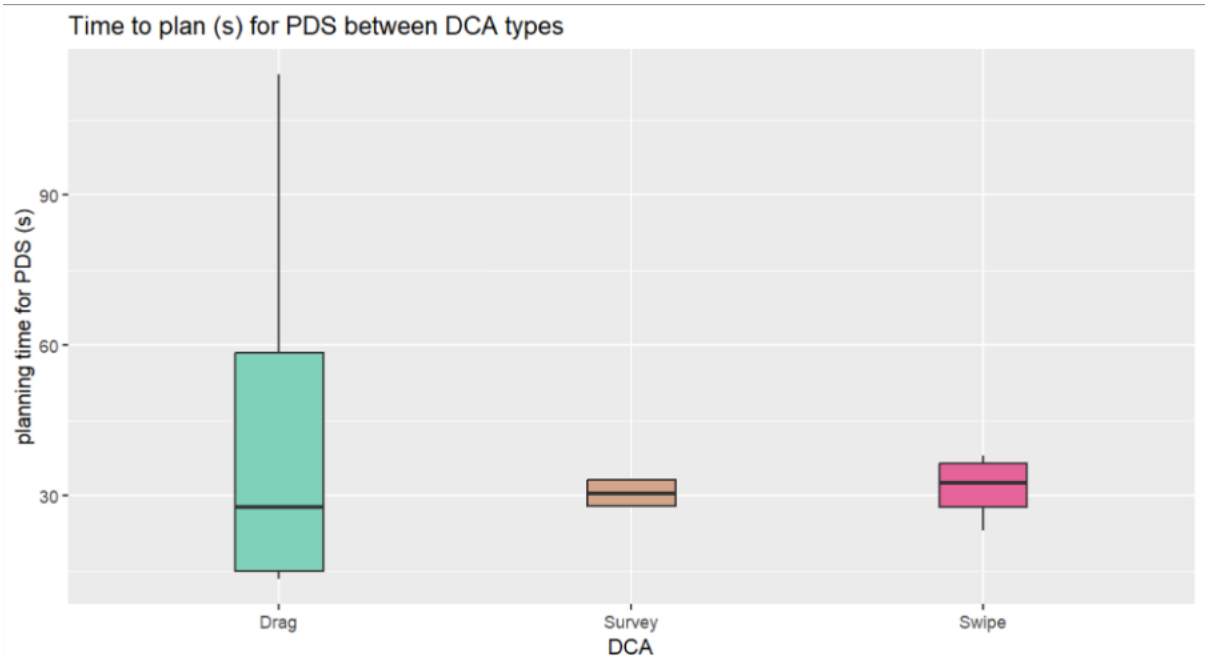

Figure 3.1: Distribution of DCA planning times by condition. Note the right-skew and high outlier for the Drag condition. No significant difference was found (p = 0.926).

4. Discussions and Limitations

4.1 Interpretation of Results

The most actionable finding from this experiment is the significant reduction in cognitive load when using the PDS compared to the BPA. This result is particularly meaningful given that planning with Google Maps is already a familiar, well-practiced task for our participants; the fact that the PDS, as a new and less familiar interface, still produced lower cognitive load suggests real merit in the route-generation concept. This is further corroborated by the qualitative data: interview participants frequently described the BPA experience as requiring active decision-making and mental effort, such as recalling campus geography or comparing options, while the PDS removed much of this burden.

The absence of a significant interaction between Task and DCA (H2) is a practically useful null result. It means no single DCA variant disproportionately drives or undermines the cognitive load benefit of the PDS. However, the qualitative data tells a more nuanced story, particularly for the Drag condition. Despite being statistically indistinguishable from Survey and Swipe in completion time, the Drag interaction produced the most negative verbal feedback across all qualitative themes. Participants consistently confused the drag gesture with the slider interaction they had just seen, producing a negative transfer effect. The Drag condition also had the highest variance in completion time and a notable outlier, suggesting that when the Drag mechanism failed participants, it failed substantially.

The map readability findings (H3) point clearly to a design gap in the PDS output. While participants reported similar satisfaction levels for both maps (no significant difference), they took meaningfully longer to read the PDS map – nearly 2.4 seconds longer on average, a large effect. Qualitative feedback attributed this to missing landmarks, unlabeled stops, and unfamiliarity with the custom map style compared to Google Maps, which participants recognized immediately from prior experience. The positive transfer advantage of Google Maps was a confound we did not fully anticipate in our design.

4.2 Qualitative Themes

Qualitative interviews yielded several consistent themes that extend and contextualize the quantitative results:

- Drag interaction confusion: Most participants assigned to the Drag condition attempted to use it as a set of individual sliders rather than performing a single fluid downward gesture. The flower icon at the top of the screen, intended as a draggable handle, was not sufficiently distinctive from the slider visual language. Participants also reported frustration at being unable to backtrack to adjust a previous answer mid-drag.
- Wording ambiguity: The label 'willingness' on one of the decision factors was flagged as unclear by participants across multiple DCA conditions. It was not evident what this factor was intended to measure – physical willingness to walk, social willingness to engage, or something else.
- Scale anchoring: For the 'time' and 'sociability' factors, several participants wanted to know what the minimum and maximum values on the scale corresponded to in concrete terms (e.g., '10 minutes' vs. '60 minutes'). The absence of anchors introduced ambiguity and may have reduced the accuracy of participants' inputs.
- Map legibility: Participants noted that the PDS map lacked sufficient contextual information. Stops along the route were not labeled, and campus landmarks were not identified, making it difficult to relate the route to their own spatial knowledge of campus.
- Swipe interaction and negative affect: Several participants noted that the swipe interface evoked associations with dating applications, which produced mild discomfort or hesitation. This was an unanticipated finding with implications for the social acceptability of the interaction paradigm.

4.3 Limitations and Threats to Validity

Several limitations constrain the generalizability and reliability of our findings:

- Sample size: With N = 12, all statistical findings have low power. According to Krejcie and Morgan's sample size formula, at least 68 participants would be required to generalize findings to the UBC student population at a 90% confidence level and 10% margin of error; a standard 95% confidence level with 5% error would require 383. Our sample is therefore not generalizable and findings must be treated as exploratory.
- Single trial per participant: Each participant completed each planning task only once. Multiple trials would have allowed outliers to be identified and reduced, and would have captured learning effects over repeated use.
- Task order carryover: The significant Task x Task Order interaction suggests that the order in which participants completed the two tasks influenced their cognitive load ratings. Participants who used the BPA first may have benefited from familiarity with the scenario when switching to the PDS, inflating the PDS's advantage or vice versa.
- Wizard-of-oz map generation: The map output shown to all participants was a fixed static image regardless of their decision factor inputs. Participants who noticed this (or suspected it) may have rated the map less favorably or engaged less authentically with the planning task.
- Experimenter presence effects: All sessions were conducted with two researchers present, which may have produced social desirability effects in questionnaire responses and interview feedback.
- Convenience sampling: All 12 participants were UBC students recruited through personal networks. This introduces selection bias and limits representativeness.

5. Conclusion

This experiment evaluated the GoodTimers walk planning prototype against a baseline of Google Maps usage, testing four hypotheses around cognitive load, DCA performance, and map readability. The key finding is that the PDS significantly reduces self-rated cognitive load relative to the BPA, a result robust across all three DCA variants. This provides preliminary evidence that the route-generation concept has value for its intended use case.

However, the experiment also identified critical design gaps. The PDS map takes significantly longer to read than Google Maps, pointing to a need for better contextual anchoring, labeled stops, and familiar map conventions. The Drag DCA, while conceptually novel and capable of capturing external interest at the demo showcase, consistently confused participants and produced the most negative qualitative feedback. Minor but important revisions are needed to the wording of decision factors and the anchoring of scales.

The results position this project at outcome (c) in the validation spectrum: the concept is still worth investigating, but serious design problems have been identified that require resolution before further validation. Future work should prioritize redesigning the drag interaction to clearly differentiate it from sliders, improving map legibility, and replicating the experiment with a larger and more representative sample. The DCA input paradigm, particularly the drag mechanism, also warrants investigation as a standalone HCI contribution independent of the walk-planning application context.

References

Müller, H., Sedley, A., & Ferrall-Nunge, E. (2014). Survey research in HCI. In J. S. Olson & W. A. Kellogg (Eds.), *Ways of knowing in HCI (pp. 229-266)*. Springer.

Krejcie, R. V., & Morgan, D. W. (1970). Determining sample size for research activities. *Educational and Psychological Measurement*, 30(3), 607-610.

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Appendix A – Statistical Output Table

Full R output from all statistical tests is reproduced below for reference.

A.1 ART ANOVA – Cognitive Load (H1 & H2)

Factor	F	DF	Df.res	p-value
Task	13.46615	1	6	0.010459 *
DCA	4.17117	2	6	0.073214 .
Task Order	1.92719	1	6	0.214414
Task : DCA	1.78087	2	6	0.247083
Task : Task Order	8.81947	1	6	0.024965 *
DCA : Task Order	5.15385	2	6	0.049805 *
Task : DCA : Task Order	0.74153	2	6	0.515484

*Table A.1: ART ANOVA output for self-rated cognitive load. * = $p < 0.05$, . = $p < 0.1$*

Parameter	eta2 (partial)	95% CI
Task	0.69	[0.19, 1.00]
DCA	0.58	[0.00, 1.00]
Task Order	0.24	[0.00, 1.00]
Task : DCA	0.37	[0.08, 1.00]
Task : Task Order	0.60	[0.08, 1.00]
DCA : Task Order	0.63	[0.00, 1.00]
Task : DCA : Task Order	0.20	[0.00, 1.00]

Table A.2: Partial eta-squared effect sizes for ART ANOVA. Threshold: 0.01 = small, 0.06 = medium, 0.14 = large.

A.2 Map Satisfaction – Wilcoxon Signed-Rank Test (H3)

Statistic	p-value	Interpretation
V = 17.5	p = 1.0	No significant difference in map satisfaction between BPA and PDS

Table A.3: Wilcoxon signed-rank test comparing map satisfaction for BPA and PDS.

A.3 Map Read Time – Paired t-test and Hedge's g (H3)

Test	Statistic	df	p-value	Mean Difference
Paired t-test (one-tailed)	t = -2.9251	11	0.006905	-2.38 seconds

Table A.4: Paired t-test comparing map read time for BPA vs. PDS. Ha: true mean difference < 0 (BPA faster).

Hedge's g	Lower CI	Upper CI	Interpretation
-1.042073	-1.9598	-0.1244	Large practical effect

Table A.5: Hedge's g effect size for map read time. Hedge's g applied due to small sample (n < 20).

A.4 DCA Planning Time – Kruskal-Wallis Test (H4)

Test	Chi-squared	DF	p-value
Kruskal-Wallis rank sum test	0.15385	2	0.926

Table A.6: Kruskal-Wallis test comparing DCA planning times across Drag, Survey, and Swipe conditions.